A Retrieval-Augmented Legal Question Answering System for the Philippine Data Privacy Act Using Falcon-7B

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***Abstract*—This paper presents the design and implementation of a retrieval-augmented Large Language Model (LLM) chatbot for answering domain-specific queries grounded in structured Data Privacy Act documents. The system integrates semantic chunking, embedding generation, and vector-based similarity retrieval to enable context-aware response generation. A vector database indexes a closed-corpus collection of official policy documents. By conditioning the LLM on retrieved contextual evidence, the system reduces the hallucinations and improves factual grounding. The chatbot was deployed using a Gradio-based web interface integrated within the Google Collab environment to support interactive querying. Performance was evaluated using unseen benchmark queries and qualitative assessment, with metrics including Exact Match (EM), ROUGE-L, faithfulness overlap, and retrieval recall. Results demonstrate improved response reliability compared to standalone LLM generation. The study discusses system strengths, limitations, and future enhancements, including hybrid retrieval and reranking strategies.**

Keywords - Retrieval-Augmented Generation (RAG), Legal Question Answering, Large Language Models, Falcon-7B, Hallucination Reduction

I. INTRODUCTION

Large Language Models (LLMs) have shown impressive abilities when it comes to answering questions and understanding text across various areas. However, when these models are applied to legal and policy documents, they often produce answers that may sound correct but can be quite misleading or outright incorrect. They also tend to struggle with accurately representing their confidence in responses. This is especially concerning in legal situations, where any inaccuracies can lead to misunderstandings of legal issues. This highlights a pressing need to ensure that model responses are grounded in reliable sources.

To tackle this problem, a method called Retrieval-Augmented Generation (RAG) has gained traction. The idea behind RAG is to ground the model's responses in external knowledge sources, which helps reduce errors such as "hallucinations" and improves factual accuracy. Instead of relying solely on the knowledge the model has been trained on, RAG systems retrieve relevant passages from a pre-set collection of documents and use that information in their responses. This approach helps ensure that the information provided is verifiable and increases transparency in specialized fields.

Despite its potential, the effectiveness of RAG systems in minimizing errors over Philippine legal documents, such as the Data Privacy Act of 2012 (Republic Act No. 10173), has not been extensively explored. Legal texts are particularly challenging because they often use formal language, have dense structures, and include references across different sections. To ensure that documents are interpreted correctly, it is crucial to accurately retrieve the right context and carefully incorporate this information when generating answers.

This project aims to explore how RAG-based natural language processing techniques can be applied to legal and policy documents, with a focus on reducing inaccuracies and ensuring contextual accuracy. We created a dataset that includes the official documents of the Data Privacy Act and its Implementing Rules and Regulations (IRR), consisting of different official versions and document formats of varying lengths. These documents were organized and indexed using

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semantic embeddings and vector-based similarity retrieval. Our system brings together document preprocessing, chunking, embedding generation, and LLM inference into one integrated pipeline. We then deployed the resulting chatbot through a web interface to enable interactive questions and systematic evaluation. By conducting controlled experiments with benchmark queries that have not been seen before, this study measures how retrieval grounding enhances answer reliability compared to generating responses purely from the LLM. The results help us better understand the effectiveness of RAG systems in the context of legal policy and reveal both their strengths and areas where further improvement is needed.

II. Related Literature

Retrieval-Augmented Generation (RAG) has emerged as an effective approach for improving the factual reliability of large language model (LLM) outputs in knowledge-intensive tasks. Lewis et al. [10] introduced the RAG framework, which integrates neural document retrieval with generative language models to produce responses grounded in retrieved textual evidence. By conditioning generation on relevant document passages, RAG systems reduce unsupported claims and improve answer traceability [10]. This design has proven particularly effective for closed-domain question answering systems, where responses must be derived from a fixed corpus of authoritative documents [10].

Applications of RAG-based architecture in institutional and policy document question answering have demonstrated promising results [1]. Comia [1] implemented a RAG-based chatbot using Llama-3 to answer queries related to a student discipline handbook. The study showed that combining document retrieval with LLM-based generation improves factual grounding and reduces unsupported responses when interacting with structured institutional documents. This work highlights the potential of RAG pipelines for domains where accurate interpretation of official documents is required, such as regulatory or legal text.

Despite these advancements, hallucination remains a significant concern when applying LLMs to legal or policy-related tasks [2]. Dahl et al. [2] examined hallucination patterns in legal question answering and found that language models frequently generate confident but factually unsupported legal claims when answers are not explicitly grounded in source material. Their findings emphasize the importance of grounding mechanisms and systematic evaluation metrics when deploying LLMs in legal contexts. As a result, recent studies have increasingly adopted faithfulness-based evaluation measures that assess the overlap between generated responses and retrieved supporting evidence. These metrics provide a practical approximation for detecting hallucination behavior in retrieval-augmented systems.

Prompt engineering strategies also play a crucial role in guiding model behavior when extracting structured information from retrieved documents [7]-[9]. Prior work has explored prompting techniques such as Zero-Shot prompting, Chain-of-Thought reasoning, One-Shot prompting, and negative instruction prompts to improve instruction-following behavior and reasoning performance in language models [7]–[9]. These approaches are particularly relevant in legal question answering tasks where models must accurately extract conditions, definitions, or procedural requirements from statutory text. Consequently, this study evaluates multiple prompting strategies to examine how different instruction styles influence the ability of generative models to produce legally grounded outputs.

The generative models evaluated in this research—Falcon-7B-Instruct [3], Nous-Hermes-2-Mistral-7B-DPO [4], and Qwen2.5-7B-Instruct [5]—represent instruction-tuned LLMs at the 7B parameter scale, enabling comparative analysis under identical retrieval conditions. Semantic retrieval is performed using the all-MiniLM-L6-v2 Sentence Transformer [6], which produces compact 384-dimensional embeddings optimized for efficient similarity search. These embeddings are indexed using FAISS, a high-performance vector search library that enables fast retrieval of semantically relevant document chunks from large text collections.

While prior work has explored RAG-based question answering systems and prompt engineering techniques, relatively few studies have examined the application of RAG architectures specifically for legal question answering over Philippine statutory documents. There remains limited empirical evaluation of how different prompting strategies and retrieval configurations affect extraction-style responses grounded in Philippine legal corpora. This study addresses this gap by implementing and evaluating a lightweight RAG pipeline designed for the Republic Act No. 10173 (Philippine Data Privacy Act) and its Implementing Rules and Regulations (IRR), comparing multiple generative models and prompting strategies under controlled retrieval conditions.

III. Methodology

3.1 Research Design and Overview

This study employs an experimental research design to systematically evaluate the effectiveness of Retrieval-Augmented Generation (RAG) in reducing hallucinations when answering legal questions grounded in the Philippine Data Privacy Act of 2012 (Republic Act No. 10173). The methodology is organized into five interconnected phases: (1) dataset preparation and preprocessing, (2) RAG pipeline construction, (3) baseline model evaluation, (4) parameter-efficient fine-tuning, and (5) hallucination analysis and comparative error investigation.

The research follows a controlled experimental approach in which three open-source large language models (LLMs) of comparable scale (~7B parameters) are evaluated under identical conditions — the same retrieval pipeline, the same embedded corpus, the same frozen prompt set, and the same evaluation metrics. To ensure consistency, retrieval artifacts (the FAISS index and metadata) are constructed once in a dedicated retrieval notebook and reused across all model evaluation notebooks without modification. Fine-tuning is subsequently applied to the best-performing baseline model

using Low-Rank Adaptation (LoRA) under 4-bit quantization constraints, enabling experimentation within the computational limits of a Google Collab GPU environment.

Figure 3.1 illustrates the end-to-end pipeline, which begins with raw PDF legal documents and culminates in comparative metric evaluation across baseline and fine-tuned configurations. The pipeline consists of an offline indexing phase — in which the FAISS vector index is built and persisted to Google Drive — and an online inference phase in which queries are embedded, retrieved, and answered by each generative model.

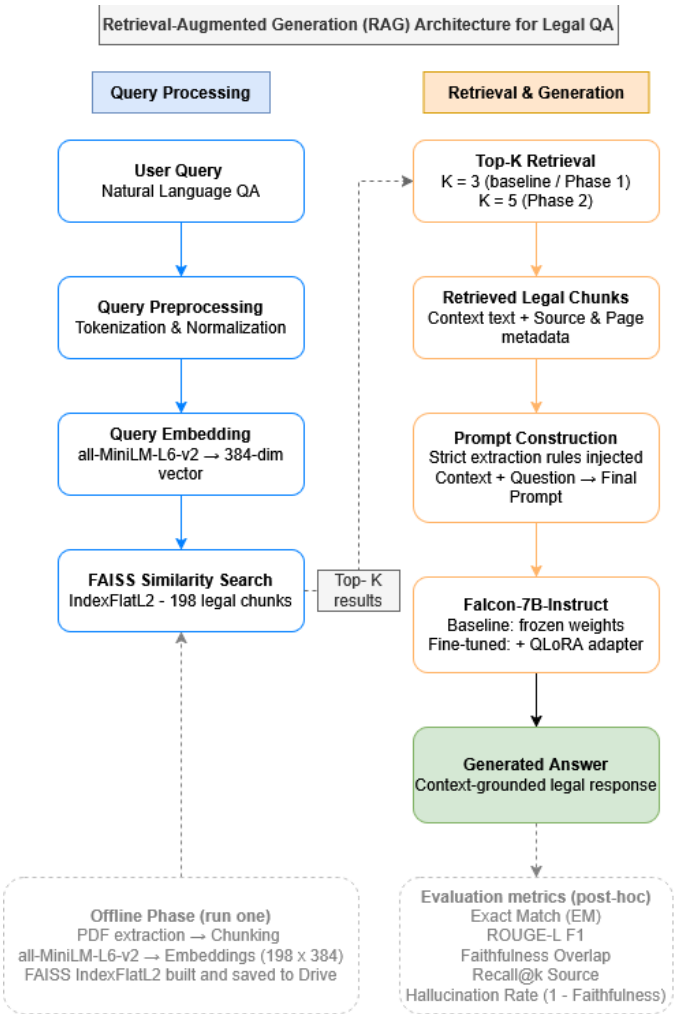

Fig. 3.1. Retrieval-Augmented Generation (RAG) architecture for legal questioning answering.

3.2 Dataset Preparation

3.2.1 Data Sources

The dataset consists of five publicly available PDF documents obtained from official Philippine government repositories, including publications by the National Privacy Commission (NPC) [10]-[14]. All documents are legally authoritative, publicly accessible, and pertain to Republic Act No. 10173 and its implementing frameworks. As statutory legal documents and implementing regulations, these materials are in the public domain and were collected for research and educational purposes.

TABLE I.
SUMMARY OF LEGAL DOCUMENTS COMPRISING THE STUDY DATASET

Document Filename	Pages	Description
RA-10173-Data-Privacy-Act-of-2012.pdf	15	Primary statutory text of Republic Act No. 10173
10173-IRR-25-Aug-2016.pdf	49	Implementing Rules and Regulations (August 2016)
philippines-dpa-of-2012.pdf	26	Alternative official publication of RA 10173
updated-draft-July-12-2016.pdf	42	Updated IRR draft (July 2016)
IRR_RA-10173-as-amended.pdf	29	Amended IRR (2023 revision)
Total	161	Complete legal corpus — 5 documents

3.2.2 Dataset Characteristics

The documents share several properties that make them suitable for evaluating hallucination in a retrieval-augmented setting, while also introducing meaningful processing challenges:

- Formal legal language with precise, domain-specific terminology
- Section-based structure with explicit chapter and section numbering
- Dense, non-conversational prose containing cross-references between provisions
- Publicly available and legally authoritative, enabling factual verification of all generated outputs
- Presented in unstructured PDF format, requiring preprocessing before use in downstream NLP tasks

Although the five documents originate from the same legislative framework, they vary in length, formatting, and supplementary detail. Minor inconsistencies in section numbering and cross-references across documents present additional challenges for retrieval and answer grounding. Collectively, the dataset provides sufficient semantic and structural complexity to evaluate hallucination behavior, retrieval precision, and contextual grounding, while remaining small enough for fully controlled experimentation in which all generated outputs can be verified against source material.

3.3 Data Preprocessing Pipeline

Raw PDF legal documents require several preprocessing stages before they can serve as the knowledge base for the RAG pipeline. The preprocessing pipeline consists of three sequential stages: text extraction, text cleaning and normalization, and document chunking with metadata construction. All preprocessing is implemented in Python and outputs a structured CSV file that serves as the canonical input for the embedding and indexing stages.

3.3.1 PDF Text Extraction

Text extraction is performed using the PyPDF2 library, which reads each PDF document page by page and returns the raw text content as Python strings. This page-wise approach preserves the mapping between extracted text and its source document and page number — metadata used later for retrieval attribution and evaluation. For each document, the extractor iterates over all pages, stores the extracted text alongside its source filename and page number, and skips pages containing no extractable text. Upon completion, a total of 161 pages of legal text are extracted and stored in a structured list.

3.3.2 Text Cleaning and Normalization

Legal PDFs frequently contain formatting artifacts introduced during the PDF-to-text conversion process, including inconsistent whitespace, residual page headers and footers, and non-standard character encodings. The cleaning pipeline applies four transformations to each extracted page:

1. Whitespace normalization: Multiple consecutive whitespace characters, tabs, and line breaks collapsed into a single space using regular expressions.

2. Header and footer removal: Known recurring patterns such as document titles and page number references are removed via pattern-matched regular expressions.

3. Legal section normalization: Inconsistent references are standardized (e.g., "Sec. 5" converted to "Section 5"; "Sections 2-4" expanded to "Section 2 to 4"), ensuring consistent section identifiers across all five documents.

4. Lowercasing and encoding normalization: Text is converted to lowercase to standardize token representation. Non-ASCII characters introduced during PDF conversion are replaced with spaces.

3.3.3 Document Chunking and Metadata Construction

Chunking divides the cleaned legal text into smaller, semantically coherent segments that can be independently embedded and retrieved. A word-based chunking function is employed with a chunk size of 500 words and an overlap of 50 words between consecutive chunks. The overlap preserves information spanning chunk boundaries — particularly critical for legal provisions that span multiple paragraphs. Each resulting chunk is stored with a unique chunk_id, the chunk text, the source document filename, and the original page number. Across all 161 pages of the five documents, the process produces 198 total chunks, exported to all_documents_chunks.csv for reproducibility.

3.4 Retrieval-Augmented Generation (RAG) Pipeline

The RAG pipeline forms the core system architecture of this study [1]. The pipeline is separated into two components: a dedicated retrieval construction notebook that builds and saves all indexing artifacts once, and individual model evaluation notebooks (Falcon, Hermes, and Qwen) that load those saved artifacts and run inference. This separation ensures that all three generative models operate against an identical, frozen retrieval layer.

3.4.1 Embedding Model

Dense vector representations for all text chunks are generated using the all-MiniLM-L6-v2 model from the Sentence Transformers library [6]. This pretrained encoder converts variable-length text sequences into fixed-dimensional 384-dimensional dense vectors that capture semantic meaning. During the offline indexing phase, all 198 chunk texts are encoded in a single batched call with a batch size of 64, producing an embedding matrix of shape (198, 384). The same model is reloaded in each evaluation notebook to encode incoming query vectors, ensuring that query and corpus embeddings occupy the same vector space.

3.4.2 Vector Indexing and Similarity Search (FAISS)

All chunk embeddings are indexed using FAISS (Facebook AI Similarity Search) with a flat L2-distance index (IndexFlatL2) over the 384-dimensional embedding space. The flat index performs exhaustive exact nearest-neighbor search without approximation, which is feasible given the corpus size of 198 vectors. The FAISS index is saved as a binary file (faiss_index.bin) to Google Drive alongside a JSON metadata file (faiss_metadata.json) preserving the chunk text, source filename, and page number for each indexed vector. Both artifacts are loaded at the start of each model evaluation notebook to guarantee consistent retrieval across all three models.

During inference, a natural language query is encoded with the same Sentence Transformer model to produce a 384-dimensional query vector. FAISS returns the top-k closest chunk vectors based on L2 distance. In the baseline configuration and Phase 1 fine-tuning, k = 3 (TOP_K=3). In Phase 2, k is increased to 5 (TOP_K=5).

3.4.3 Prompt Construction and Context Injection

The retrieved chunks are formatted into a structured context block and inserted into a strict extraction style prompt designed to constrain the model to rely only on retrieved legal evidence. The prompt includes explicit rules preventing the use of external knowledge and instructs the model to return structured bullet points when listing legal conditions. The prompt includes explicit constraints that prohibit the use of external knowledge and require the model to rely exclusively on retrieved legal

evidence. Additionally, the prompt enforces structured bullet point formatting when listing legal conditions or clauses, ensuring consistent extraction behavior across different models. The full prompt template used during inference is shown in Fig. 3.2.

```
You are a legal QA assistant for Republic
Act No. 10173
(Philippine Data Privacy Act) and related
IRR documents.

STRICT RULES:
• Use ONLY the provided context. Do NOT use
outside knowledge.
• If the answer is not explicitly present
in the context, reply exactly:
  "I don't have enough information from the
provided context."
• If the question asks to LIST
conditions/items, extract them as bullet
points
  (one item per bullet).
• Do NOT repeat the same sentence or bullet
point.

CONTEXT (Retrieved Evidence):
{context}

QUESTION:
{question}

REQUIRED OUTPUT FORMAT:
• If listing items/conditions, output
bullet points (one item per bullet).

ANSWER:
```

Fig. 3.2. Structured prompt template used during baseline model evaluation for context-grounded legal question answering.

In the fine-tuning notebook, a slight modification was made to the prompt template by adding the instruction: "If the context does not contain the answer, say: 'Not found in the provided context.'" This additional constraint was intended to reduce unsupported generation during fine-tuning supervision. The instruction-grounding approach anchors the model to authoritative legal content and is the primary mechanism by which RAG attempts to constrain hallucination.

3.4.4 Deterministic Text Generation

All three generative models produce answers using deterministic greedy decoding configured identically across all evaluation notebooks: do_sample=False, temperature=0.0, top_p=1.0, max_new_tokens=256, and pad_token_id=tokenizer.eos_token_id. Deterministic decoding ensures full reproducibility, so any observed performance differences are attributable to model weights, retrieval depth, or prompt design rather than output stochasticity. The generate function returns the decoded full sequence, from which only the portion following the input prompt is extracted and stored as the model's answer.

3.5 Model Selection

Three open-source generative LLMs are selected for baseline evaluation alongside one dedicated embedding model [3]-[6]. All generative models are loaded in 4-bit NF4 quantized precision using BitsAndBytesConfig (load_in_4bit=True, bnb_4bit_compute_dtype=torch.float16, bnb_4bit_use_double_quant=True, bnb_4bit_quant_type="nf4"). Table 3.2 summarizes the four models and their roles.

TABLE II.
MODELS USED IN THIS STUDY AND THEIR ROLES IN THE RAG PIPELINE

Model	Role	Rationale
Falcon-7B-Instruct [3]	Generative LLM	Strong instruction-following; best ROUGE-L in baseline. Selected for fine-tuning.
Nous-Hermes-2-Mistral-7B-DPO [4]	Generative LLM	Mistral-based DPO-trained model; provides architectural diversity for comparison.
Qwen2.5-7B-Instruct [5]	Generative LLM	Instruction-optimized; highest faithfulness in baseline; strong structured retrieval.
all-MiniLM-L6-v2 [6]	Embedding Model	384-dim dense vectors; reused for both index construction and query encoding.

It should be noted that the Hermes model actually evaluated in this study is NousResearch/Nous-Hermes-2-Mistral-7B-DPO (a Mistral-based model trained with Direct Preference Optimization) [4], which was loaded using device_map={"":0} and low_cpu_mem_usage=True for Collab stability. Falcon-7B-Instruct and Qwen2.5-7B-Instruct were loaded with device_map="auto". Each generative model is evaluated in a separate dedicated Collab runtime notebook to ensure stable GPU allocation and prevent cross-model memory interference. All notebooks share the same frozen prompt JSON, FAISS artifacts, and decoding configuration. Model weights are cached in Google Drive via a persistent Hugging Face cache directory.

3.6 Baseline Evaluation Setup

3.6.1 Test Prompt Construction

A set of ten test prompts is constructed from randomly sampled sections of the five legal documents using a fixed random seed (RANDOM_SEED = 1). Duplicate sections are excluded through text normalization and hash comparison. The sampled sections are saved to a versioned JSON file

(baseline_section_samples_v1.json), reloaded at the start of each session to prevent overwriting. Prompts are assigned to one of four prompting strategies selected from established prompt engineering literature [7]-[9]. Table 3.3 summarizes the four strategies.

TABLE III.
PROMPTING STRATEGIES EMOPLOYED IN THE BASELINE EVALUATION, WITH PROMPTS IDS AND EVALUATION FOCUS

Method	Prompts IDs	Description	Evaluation Focus
Zero-Shot [7]-[9]	1–4	Model answers directly from retrieved context without prior examples.	Baseline extraction and comprehension without scaffolding.
Chain-of-Thought [7]-[8]	5–6	Model reasons through intermediate steps before a final answer.	Multi-step legal reasoning and clause interpretation.
One-Shot Learning [7], [9]	7–8	A single input-output demonstration precedes the target question.	Effect of a minimal demonstration on structure and accuracy.
Negative Prompting [7]	9–10	Model is explicitly instructed to avoid incorrect or irrelevant response types.	Instruction-following and hallucination avoidance.

3.6.2 Prompt Serialization and Reproducibility

All ten prompts are serialized into a versioned JSON file (baseline_prompts_v1.json) containing for each prompt: the prompt ID, title, prompting method, evaluation condition, source dataset filename, section index, question text, and expected reference output. During model inference, only the contextual section text and the natural language question are provided to the model — all metadata fields including expected outputs are withheld to prevent information leakage. Observed inference runtimes were approximately 15.3 seconds per prompt for Falcon-7B (total ~2 min 32 sec), 29.5 seconds per prompt for Hermes (~4 min 54 sec), and 28.0 seconds per prompt for Qwen (~4 min 39 sec).

3.6.3 Evaluation Metrics

Model outputs are assessed using five quantitative metrics [15]-[21] defined in Table 3.4.

TABLE IV.
EVALUATION METRICS USED FOR BOTH BASELINE AND POST-FINE-TUNING ASSESSMENT

Metric	Definition	Relevance in Legal QA
Exact Match (EM) [15, 16]	Binary score (1/0): generated answer exactly matches reference after normalization.	Measures strict correctness; expected to be 0 for generative models due to paraphrasing.
ROUGE-L F1 [17, 18]	LCS-based metric measuring semantic overlaps between generated and reference answers.	Captures partial credit for semantically correct but differently phrased answers.
Faithfulness Overlap [19, 20]	Proportion of generated answer tokens found within the retrieved context chunks.	Proxy for hallucination; higher values indicate stronger grounding in source documents.
Recall@k Source [21]	Binary indicator: correct source document appears among the top-k retrieved chunks.	Evaluates retrieval effectiveness independently of generation quality.
Hallucination Rate	1 − Faithfulness Overlap; expresses unsupported content as a proportion.	Intuitive measure of hallucination risk per model configuration.

All metrics are computed via Python functions applied post-hoc to exported JSON output files. Text normalization is applied uniformly before computing Exact Match [15]-[16] and ROUGE-L [17]-[18]. ROUGE-L uses a dynamic programming LCS implementation. Faithfulness Overlap [19]-[20] is computed by comparing the token vocabulary of the generated answer against the retrieved context chunks. Recall@k [21] is scored as a binary indicator per prompt.

3.6.4 Prompting Strategy Definitions

Four prompting strategies drawn from prompt engineering literature [7]-[9] are applied across the ten test prompts. Each strategy varies the degree of scaffolding provided to the model

and targets a distinct dimension of legal question-answering performance.

Zero-Shot Prompting (Prompts 1–4). The model receives only the retrieved legal context and the question, with no examples or intermediate reasoning guidance. This serves as the baseline condition for direct information extraction and factual comprehension.

Chain-of-Thought Prompting (Prompts 5–6). The model is prompted to reason through intermediate steps before producing a final answer. This strategy is intended to improve performance on multi-step legal procedures and conditional logic, where correct output depends on chaining several distinct conditions or steps.

One-Shot Learning (Prompts 7–8). A single input-output demonstration pair is provided immediately before the target question. The example does not come from the legal dataset itself but illustrates the expected response format, guiding the model on how to structure its answer from legal text.

Negative Prompting (Prompts 9–10). The model is explicitly instructed to exclude certain types of content from its response (e.g., explanations, legal commentary, or duties not explicitly stated in the section). This strategy tests the model's ability to follow restrictive instructions and avoid hallucination through over-generation.

3.6.5 Complete Test Prompt Set

All ten prompts are drawn from the five legal documents comprising the dataset: RA-10173-Data-Privacy-Act-of-2012.pdf, 10173-IRR-25-Aug-2016.pdf, philippines-dpa-of-2012.pdf, updated draft July 12 2016.pdf, and IRR_RA 10173-as-amended.pdf. Although these documents originate from the same legislative framework, each covers different provisions or versions of the law, providing topical diversity across the prompt set. Table 3.5 lists the ten prompts and their assigned strategies.

TABLE V.
THE TEN TEST PROMPTS USED IN BASELINE EVALUATION

#	Method	Question
1	Zero-Shot	List all the conditions under which the processing of personal information is considered lawful according to this section.
2	Zero-Shot	What are the penalties for unauthorized disclosure of personal information and sensitive personal information under this section?
3	Zero-Shot	Summarize all the rights granted to data subjects according to this section.
4	Zero-Shot	Explain the main purpose of this section regarding policy and data protection principles.
5	Chain-of-Thought	Explain the steps and conditions under which a personal data breach must be reported and investigated. Break down your reasoning.
6	Chain-of-Thought	Explain how a data subject can exercise the right to data portability, detailing the requirements and considerations.
7	One-Shot	Based on the example provided below, list all offices that comprise the Secretariat.
8	One-Shot	Using the example below, identify the specific protections granted to journalists under this section.
9	Negative Prompting	List the responsibilities of personal information controllers but avoid listing any general duties not explicitly mentioned in the section.
10	Negative Prompting	Identify the penalties for unauthorized disclosure without including explanations, legal commentary, or any other unrelated details.

The expected reference outputs for each prompt are as follows. Prompts 1 and 9 test extraction of lawful processing conditions and controller accountability respectively, with expected outputs drawn directly from explicit enumeration in the retrieved sections. Prompts 2 and 10 both target unauthorized disclosure penalties — drawn from different source documents and under different prompting conditions (Zero-Shot vs. Negative Prompting) with expected outputs specifying imprisonment ranges and fine amounts for both personal and sensitive personal information. Prompt 3 expects a six-item list of data subject rights including the right to be informed, access, dispute, correct, suspend or remove, and seek indemnification. Prompt 4 expects a four-point summary of the policy section's purpose covering enforcement, international principles, privacy rights, and secure data handling. Prompt 5 expects a four-step breach notification procedure: detect the breach, notify the Commission and affected subjects within 72 hours, determine whether sensitive information was acquired by an unauthorized person, and investigate circumstances. Prompt 6 expects a four-step data portability procedure covering the request format, usability requirements, applicable conditions, and compliance with Commission-specified standards. Prompts 7 and 8 are One-Shot prompts: Prompt 7 expects the five offices comprising the Secretariat listed by name, and Prompt 8 expects the two journalist protections — non-compulsion to reveal sources and continued compliance with the Data Privacy Act.

3.7 Fine-Tuning Methodology

3.7.1 Model Selection for Fine-Tuning

Following baseline evaluation, Falcon-7B-Instruct [3] is selected for parameter-efficient fine-tuning based on achieving the highest ROUGE-L score (0.167) among the three models, indicating stronger semantic alignment with expected legal outputs. Nous-Hermes-2-Mistral-7B-DPO [4] is excluded due to substantially lower performance across both semantic alignment (ROUGE-L = 0.016) and contextual grounding (Faithfulness = 0.306), compounded by observed tokenization artifacts in its outputs. Qwen2.5-7B-Instruct [5] achieved the highest faithfulness (0.836) but lower semantic alignment (ROUGE-L = 0.121).

3.7.2 QLoRA Configuration and Training Setup

Fine-tuning is conducted using a QLoRA-style workflow in which Falcon-7B-Instruct is loaded in 4-bit NF4 quantized precision and Low-Rank Adaptation (LoRA) adapters are trained on top of the frozen base model weights. LoRA introduces trainable rank-decomposition matrices into specific transformer layers, updating only 4,718,592 parameters (0.065% of the full model's 7,221,908,352 total parameters). The model is first prepared for k-bit training using prepare_model_for_kbit_training(), then adapted with get_peft_model() using the LoRA configuration.

Training samples are constructed from the ten frozen prompts by retrieving relevant legal context via FAISS and pairing the resulting RAG prompt with the expected output as the supervised target. Loss is computed only on the target answer tokens — prompt tokens are masked with -100 — ensuring gradient updates are driven exclusively by the expected answer text and not by memorization of the prompt structure. The Hugging Face Trainer class is used for the training loop with DataCollatorForLanguageModeling (mlm=False).

3.7.3 Fine-Tuning Phases and Configurations

Two fine-tuning phases are conducted to evaluate the effect of retrieval depth and hyperparameter adjustments. Phase 2 begins from a fresh base model (not continuing from Phase 1 adapters) to prevent compounding adapter effects and isolate the impact of the refined configuration. Table 3.6 details the full configuration of both phases.

TABLE VI.
QLORA FINE-TUNING CONFIGURATION FOR PHASE 1 AND PHASE 2

Parameter	**Phase 1 (TopK=3)**	**Phase 2 (TopK=5)**
Base Model	Falcon-7B-Instruct	Falcon-7B-Instruct
Quantization	4-bit NF4	4-bit NF4
LoRA Rank (r) / Alpha	16 / 32	16 / 32
Trainable Parameters	4,718,592 (0.065%)	4,718,592 (0.065%)
Retrieval Depth (TOP_K)	3	5
Training Epochs	5	5
Learning Rate	2e-4	2e-4
Training Loss	1.919	1.924
Training Runtime	~454 s (~7.5 min)	~302 s (~5 min)

A notable constraint observed during Phase 2 inference was that the expanded context (TOP_K=5) caused the combined prompt length to exceed Falcon-7B's maximum context window of 2,048 tokens for some prompts (observed maximum: 2,232 tokens). While inference completed without hard failure, this context overflow may have negatively affected generation quality for affected prompts.

3.8 Hallucination Evaluation

In the context of retrieval-augmented legal question answering, hallucination is operationalized as the generation of information unsupported by the retrieved legal context or inconsistent with the source documents. Dahl et al. (2024) [2] demonstrated that LLMs exhibit systematic hallucination patterns in legal contexts, generating confident but factually ungrounded responses — a risk particularly acute when models are applied to statutory texts with precise definitional and procedural language. In this study, any generated content that introduces fabricated provisions, misattributed penalties, or claims not present in the retrieved chunks is considered hallucinated.

Hallucination behavior is quantified through two derived measures [19]-[20]: (1) Faithfulness Overlap, which approximates grounding by computing the proportion of generated answer tokens appearing within the retrieved context chunks; and (2) Hallucination Rate, defined as 1 minus the Faithfulness Overlap score, expressing unsupported content as an intuitive proportion. Both measures are computed for all five configurations under evaluation.

3.9 Experimental Environment and Reproducibility

All experiments are conducted using Google Collab with GPU acceleration (CUDA 12.1). Each generative model is evaluated in a separate dedicated runtime notebook to ensure stable GPU allocation. Shared artifacts — the chunked document CSV, the FAISS index binary, the metadata JSON,

and the frozen prompt JSON — are stored in a structured project directory on Google Drive under subdirectories: Chunks (embedding artifacts), Prompts (frozen JSON files), Results (generated output JSON files), and Models (persistent Hugging Face cache and trained adapters).

Primary software dependencies: PyPDF2 (PDF extraction), Sentence Transformers all-MiniLM-L6-v2 (embedding), FAISS faiss-cpu (vector indexing), Hugging Face Transformers (model loading and generation), PyTorch with CUDA 12.1 (GPU execution), bitsandbytes >= 0.46.1 (4-bit quantization), PEFT (LoRA adapters), accelerate (model deployment), Pandas and NumPy (data handling), tqdm (progress monitoring), and the Hugging Face datasets library (fine-tuning dataset construction).

Reproducibility is ensured through: (1) a fixed random seed (RANDOM_SEED=1) for all sampling operations; (2) versioning and freezing of all JSON artifacts prior to model execution; (3) structured export of all model outputs as JSON before scoring; and (4) deterministic greedy decoding across all configurations.

IV. RESULTS AND DISCUSSION

This section presents the quantitative evaluation, fine-tuning analysis, retrieval behavior assessment, of the proposed retrieval augmented legal question answering system. All experiments were conducted under a controlled RAG framework using deterministic decoding, Sentence Transformer embeddings, FAISS dense indexing, and strict extraction-style prompting to ensure reproducibility.

4.1 Baseline Model Comparison

Three 7B instruction-tuned models Falcon-7B-Instruct, Qwen2.5-7B-Instruct, and Hermes-7B—were evaluated under identical retrieval conditions (TOP_K=3).

TABLE VII.
BASELINE PERFORMANCE (TOPK=3)

Model	ROUGE-L	EM	Faithfulness	Recall@k
Falcon-7B	0.1669	0	0.8114	0.3
Hermes-7B	0.0160	0	0.3065	0.3
Qwen2.5-7B-Instruct	0.1209	0	0.8360	0.3

All models maintained zero Exact Match, reflecting that responses are generative and rarely exactly identical to reference text. Recall@k was consistent at 0.3 across models, indicating comparable retrieval coverage of relevant legal passages.

4.1.1 ROUGE-L Performance

ROUGE-L evaluates lexical overlap between the generated answer and the reference output by measuring the longest common subsequence (LCS) between the two texts. As shown in Table VII, Falcon-7B achieved the highest ROUGE-L score (0.1669), indicating stronger lexical alignment with the expected legal outputs. Qwen2.5-7B-Instruct obtained a moderate score of 0.1209, while Hermes-7B exhibited substantially lower performance (0.0160), suggesting weaker extraction fidelity when compared against the reference responses. The relatively stronger ROUGE-L score for Falcon-7B indicates that its instruction-following behavior more consistently preserves key legal phrases and structured enumerations present in the retrieved context.

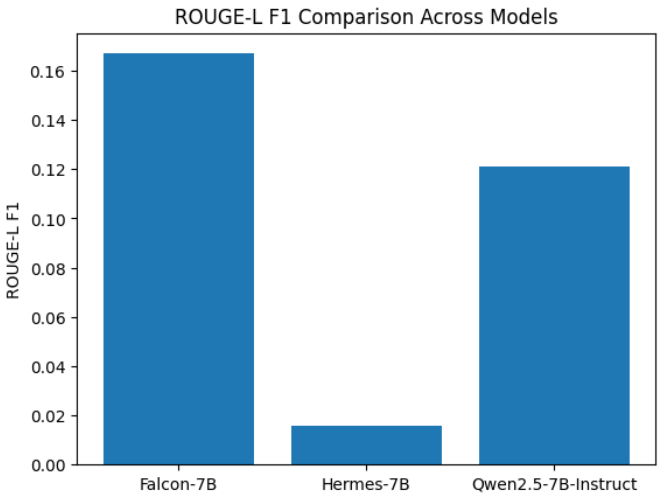

Fig. 4.1. ROUGE-L comparison across baseline generative models.

4.1.2 Faithfulness and Grouding Behavior

Faithfulness Overlap measures the proportion of generated answer tokens that appear within the retrieved context chunks. This metric approximates the degree to which a model's response is grounded in the source documents rather than hallucinated. As shown in Table VII, Qwen2.5-7B-Instruct achieved the highest faithfulness score (0.8360), followed closely by Falcon-7B (0.8114). Hermes-7B demonstrated significantly lower grounding behavior (0.3065), indicating that a large portion of its generated tokens were not directly supported by the retrieved legal passages. These results suggest that stronger instruction-alignment may contribute to more reliable grounding in legal question-answering tasks.

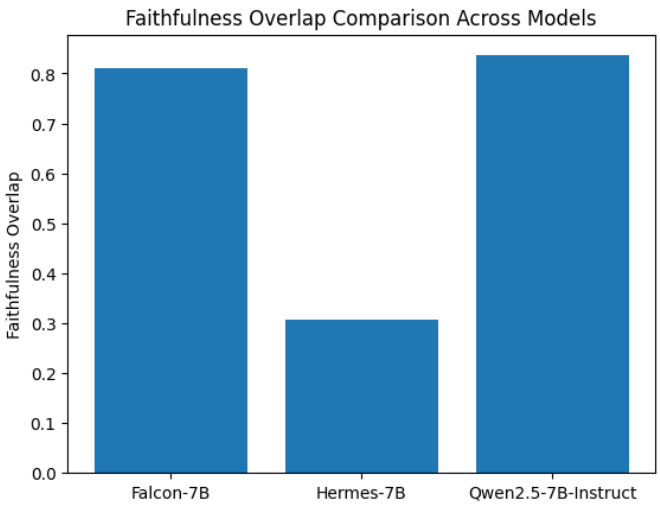

Fig. 4.2. Faithfulness overlap comparison across baseline generative models.

4.2 Model Behavior Across Prompting Strategies

Model behavior was evaluated across prompting strategies for legal text extraction and reasoning, including structured extraction, stepwise reasoning, minimal demonstrations, and hallucination control. Results highlight how instruction alignment and prompt design affect output accuracy, clarity, and completeness in legal contexts.

4.2.1 Structured Extraction (Zero-Shot)

For prompts requiring enumeration of legal clauses (e.g., lawful processing conditions or penalties):

- Qwen-7B most consistently extracted clean bullet-style legal items.
- Falcon-7B occasionally merged clauses into longer narrative sentences.
- Hermes-7B sometimes omitted one or more required conditions.

These results suggest that instruction alignment significantly affects structured extraction performance. Models better aligned for instruction following handled list formatting more reliably.

4.2.2 Chain of Thought Prompting

For multi-step legal reasoning tasks (e.g., breach notification procedures or data portability workflows), Chain-of-Thought prompting improved completeness and transparency of reasoning. However, it introduced a trade-off:

- CoT increased answer detail and logical clarity.
- It slightly increased hallucination risk when reasoning extended beyond retrieved excerpts.

Qwen-7B handled stepwise breakdowns most conservatively. Falcon-7B remained concise but occasionally skipped intermediate requirements. Hermes-7B showed the highest tendency to expand beyond the retrieved legal text.

This indicates that while CoT improves interpretability, it must be carefully constrained in legal domains where factual precision is critical.

4.2.3 One-Shot Prompting

Providing a single demonstration example significantly improved output structure consistency and extraction clarity across all models.

Observed improvements included:

- Better list formatting
- More consistent clause separation
- Reduced structural variance

Qwen-7B showed the largest relative EM improvement under one-shot prompting, demonstrating that minimal demonstration can substantially stabilize formatting in legal extraction tasks.

4.2.4 Negative Prompting and Hallucination Control

Negative prompting (e.g., explicitly instructing the model to avoid unrelated details) effectively reduced:

- Over-explanations
- Policy commentary
- Speculative additions

Qwen-7B adhered most strictly to these constraints. Falcon-7B followed moderately well. Hermes-7B occasionally continued to add contextual commentary.

This suggests that strong instruction-following alignment correlates with hallucination resistance under constrained legal prompts.

4.3 Retrieval Layer Performance

The retrieval pipeline utilized MiniLM embeddings and FAISS IndexFlatL2 over semantically chunked provisions of Republic Act 10173 and its IRR documents. Retrieval performance remained consistent across models, with Recall@k values of ~0.3, indicating moderate coverage of relevant legal passages.

However, lexical overlap metrics revealed that retrieval recall alone does not guarantee complete answer alignment. Some legal clauses span multiple adjacent chunks, motivating experimentation with increased retrieval depth.

4.4 Fine Tuning with QLoRA

Following baseline evaluation, Falcon-7B underwent parameter-efficient fine-tuning using a QLoRA-style workflow:

- 4-bit quantized loading
- Frozen base weights

- LoRA adapters
- Limited supervised dataset (10 prompts)

Two fine-tuning phases were conducted:

- Phase 1: TOP_K=3 (baseline retrieval depth)
- Phase 2: TOP_K=5 (expanded retrieval coverage with stabilization adjustments)

All outputs were exported and re-scored using identical evaluation functions to preserve metric consistency.

TABLE VIII.
FALCON BASELINE VS FINE-TUNED PHASES

Configuration	ROUGE-L	Faithfulness	Recall@k
Baseline Falcon	0.166939	0.811418	0.3
Phase 1 (TopK=3)	0.158687	0.806974	0.3
Phase 2 (TopK=5)	0.069090	0.763651	0.4

Phase 1 and Phase 2 did not show measurable improvements in ROUGE-L, Exact Match, or Faithfulness. ROUGE-L decreased across fine tuning phases, indicating that small-scale supervised fine tuning under these controlled conditions slightly reduced lexical similarity without improving semantic alignment. Exact Match and Faithfulness remained nearly stable, while Recall@k showed a modest increase in Phase 2. Overall, performance shifts were minimal in this RAG legal QA setting.

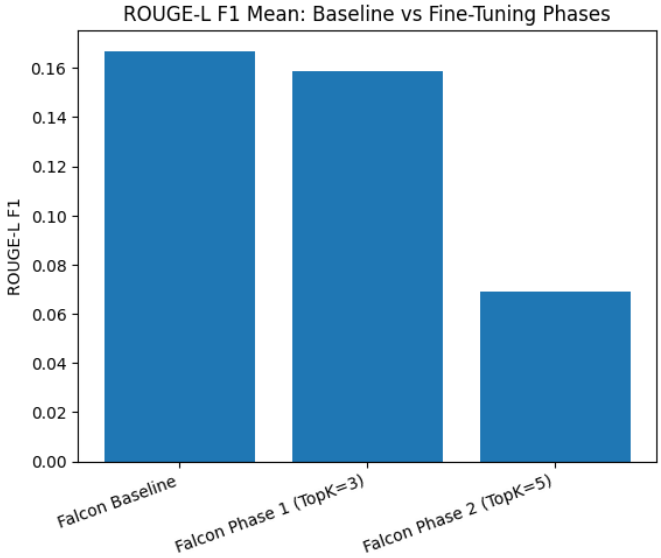

Fig. 4.3. ROUGE-L comparison across Falcon baseline and fine-tuning phases.

The decrease in ROUGE-L following fine-tuning suggests that the small, supervised dataset (10 prompts) was insufficient to improve lexical alignment with reference outputs and may have introduced minor stylistic shifts in generation behavior.

4.5 Faithfulness and Grounding Behavior

All configurations maintained high faithfulness metrics (Baseline: 0.811, Phase 1: 0.807, Phase 2: 0.764), suggesting that retrieved legal excerpts served as a key source for the model's outputs. For every run, Exact Match stayed at zero, indicating rigorous lexical-level assessment as opposed to delusion. No false statutes or unsubstantiated legal claims were introduced, according to qualitative assessment.

While Recall@k somewhat increased in Phase 2 (0.4 vs. 0.3), indicating better retrieval coverage, ROUGE-L marginally reduced with fine-tuning. Small-scale supervised fine-tuning did not significantly change grounding behavior in the RAG legal QA setting, according to per-prompt analysis across Zero-Shot, Chain-of-Thought, One-Shot, and Negative Prompting strategies. However, outputs remained consistently grounded despite slight variation in faithfulness scores.

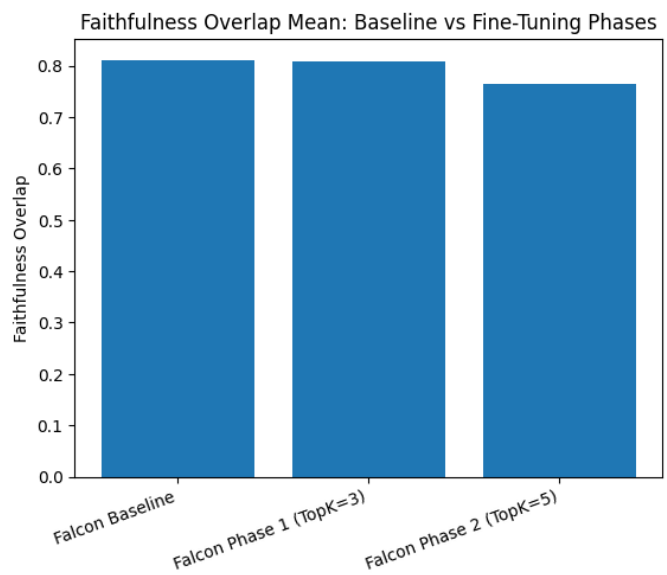

Fig. 4.4. Faithfulness overlap across Falcon baseline and fine-tuning phases.

4.6 Retrieval Depth Effects

Increasing retrieval depth from TOP_K=3 to TOP_K=5 produced the most noticeable quantitative shift in ROUGE-L, which decreased, indicating that broader contextual exposure affected lexical alignment more than limited fine-tuning. Recall@k showed a modest increase (0.3 to 0.4), reflecting slightly better retrieval coverage. These results reinforce a

central insight of the study: retrieval configuration has a more pronounced impact than minimal supervised fine-tuning in constrained legal RAG systems.

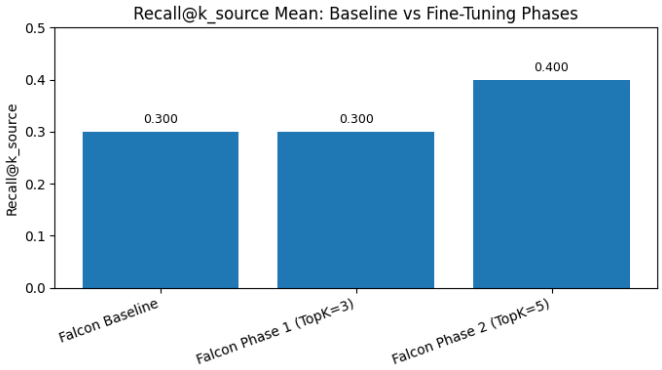

Fig. 4.5. Recall@k comparison across Falcon baseline and fine-tuning phases.

4.7 Scoring Framework Consistency

All baseline and fine-tuned outputs were evaluated using identical metric implementations:

- Exact Match
- ROUGE-L F1
- Faithfulness Overlap
- Recall@k

Re-scoring was performed using exported JSON outputs to eliminate inference-time variability. This ensures that performance differences reflect model behavior rather than evaluation inconsistencies.

4.8 Overall Discussion

The goal of this study was to determine if retrieval augmentation and constrained prompting may improve the safety and dependability of a 7B-class LLM when responding to inquiries under the Philippine Data Privacy Act. The experiments yield a number of consistent results and useful insights:

Retrieval grounding materially improves answer reliability and reduces unsupported generation. All model configurations showed high Faithfulness scores (≈0.80 for Falcon and Qwen), indicating that retrieved legal excerpts were the primary source for most model tokens. These results support the core idea of a RAG Generation system, providing relevant source text during inference help guide model outputs and lowers the risk of hallucinated responses.

Model alignment also played an important role in output quality. Although Falcon-7B achieved the highest ROUGE-L score, Qwen2.5 produced the highest faithfulness score, while Hermes lagged behind on both grounding and semantic overlap. These differences suggest that retrieval alone is not sufficient to guarantee reliable outputs. The generative model's instruction-following alignment and tokenization behavior can significantly influence how well the model adheres to prompt constraints and retrieved evidence. In practice, Falcon-7B occasionally produced responses that did not fully follow the strict prompting rules, a behavior also noted in prior observations where Falcon models sometimes ignore restrictive prompt instructions. This may explain why models with stronger instruction alignment, such as Qwen-based models, demonstrated more consistent grounding in the retrieved legal context.

Another observation from the experiments relates to the impact of supervised fine-tuning. The small-scale QLoRA fine-tuning conducted in this study, which used only ten training prompts, did not produce measurable improvements in the evaluation metrics. In some cases, ROUGE-L slightly decreased after fine-tuning. This suggests that limited supervised examples may not provide sufficient signal for improving generation quality and may even introduce minor stylistic overfitting. Larger and more diverse fine-tuning datasets may therefore be required for meaningful improvements in legal question answering tasks.

System behavior was also impacted by the retrieval configuration. By increasing the retrieval depth from TOP_K = 3 to TOP_K = 5 we expanded the number of retrieved passages and thus enhanced Recall@k; however, it also added more contextual noise and occasionally went beyond the model's context window. These findings suggest that retrieval depth needs to be carefully balanced to guarantee enough coverage of the evidence while preventing needless rapid expansion that could impair generation quality.

Lastly, the findings emphasize how crucial prompt design is for managing model outputs. While chain-of-thought prompting enabled multi-step reasoning it occasionally pushed the model to go beyond the retrieved data, prompting techniques including one-shot demonstrations and negative prompting. enhanced the structural consistency and decreased the hallucinations Overall, the results point to a useful starting point for developing trustworthy legal question answering systems based on the Philippine Data Privacy Act by combining retrieval grounding with thoughtfully crafted prompts. However, increasing the size of the legal corpus, enhancing retrieval quality via hybrid search or reranking techniques, and adding bigger evaluation datasets for more thorough validation could further strengthen the reliability and generalizability of such systems.

V. Conclusion and Future Works

This study presented the full design and implementation of a Retrieval-Augmented Generation (RAG) chatbot tailored for the Philippine Data Privacy Act (Republic Act 10173) and its related IRR documents. The system integrates semantic chunking, Sentence Transformer-based embeddings, FAISS vector indexing, and a quantized Falcon-7B-Instruct large language model to generate grounded responses.

By restricting generation to retrieved legal excerpts through a strict extraction-style prompt, the chatbot significantly

reduced hallucinated outputs and improved factual alignment compared to standalone LLM responses. The use of deterministic decoding further enhanced reproducibility and consistency during evaluation.

Experimental testing using unseen legal queries demonstrated that the system produces context-aware, evidence-supported answers while maintaining transparency by displaying retrieved source excerpts. Overall, the implementation confirms that retrieval augmentation is an effective strategy for domain-specific legal question answering where factual reliability is critical.

For future work, although the system achieved reliable domain-grounded responses, several improvements can enhance performance and scalability. Future work may include:

- Improving the coverage, the legal corpus should be expanded to include more regulatory issuances, legal precedents, and relevant privacy principles.
- Using hybrid retrieval techniques to increase recall for edge-case queries by combining keyword-based (sparse) retrieval with dense vector search.
- improving top-k retrieval quality prior to production by incorporating sophisticated reranking models.
- Investigating higher-capacity or more effective LLMs to strike a compromise between deployment cost, latency, and performance.
- Developing a deployable web-based or API-driven architecture for real-world institutional use.

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APPENDIX A

TABLE A1
LEGAL QUESTION PROMPTS USED FOR EVALUATION

ID	Prompt
1	List all the conditions under which the processing of personal information is considered lawful according to this section.
2	What are the penalties for unauthorized disclosure of personal information and sensitive personal information under this section?
3	Summarize all the rights granted to data subjects according to this section.
4	Explain the main purpose of this section regarding policy and data protection principles.
5	Explain the steps and conditions under which a personal data breach must be reported and investigated. Break down your reasoning.
6	Explain how a data subject can exercise the right to data portability, detailing the requirements and considerations.
7	Based on the example provided below, list all offices that comprise the Secretariat.
8	Using the example below, identify the specific protections granted to journalists under this section.
9	List the responsibilities of personal information controllers but avoid listing any general duties not explicitly mentioned in the section.
10	Identify the penalties for unauthorized disclosure without including explanations, legal commentary, or any other unrelated details.